

Xiao Ma

Research Scientist @ ByteDance Seed

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BIO

I'm a research scientist at ByteDance Seed, leading Seed GR robotics team. My research spans across robotics, reinforcement learning, and multi-modal models. My long-term goal is to develop intelligent embodied agents that perceive, interact with, and adapt to unstructured physical environments in a shared autonomy with humans. My work has been selected as the **best system paper finalist** and **best student paper finalist** at RSS 2019.

EDUCATION

National University of Singapore (NUS)	Aug. 2017 - Dec. 2021
Doctor of Philosophy in COMPUTER SCIENCE	
Advisor: Prof. David Hsu	
Shanghai Jiao Tong University (SJTU)	Sept. 2013 - July 2017
Bachelor of Engineering in COMPUTER SCIENCE AND TECHNOLOGY	

WORK EXPERIENCES

Principal Research Scientist at ByteDance Seed	July 2024 - present
Lead Researcher at Dyson Robot Learning Lab	Feb. 2023 - July 2024
Research Scientist at SEA AI Lab	Jul. 2021 - Jan. 2023
Research Intern at SEA AI Lab (hosted by Dr. Min Lin and Dr. Shuicheng Yan)	Apr. 2021 - Jun. 2021
Research Intern at SenseTime Research (hosted by Dr. Shuai Yi)	Oct. 2019 - Sept. 2020
Software Engineer Intern at Intel Asia Pacific R & D Center	May 2016 - Dec. 2016

TECH REPORTS

1. ByteDance Seed GR Team. "GR-Dexter Technical Report", *Technical Report*, 2025
2. ByteDance Seed GR Team. "GR-RL: Going Dexterous and Precise for Long-Horizon Robotic Manipulation", *Technical Report*, 2025
3. ByteDance Seed GR Team. "GR-3 Technical Report", *Technical Report*, 2025

PREPRINTS

1. Peiyan Li, Yuze Zhu, Yixiang Chen, Qisen Ma, Yuan Xu, Jiabing Yang, He Guan, Yan Huang, Hongtao Wu, **Xiao Ma**, Tao Kong, Liang Wang, Tieniu Tan. "BridgeVLA++: A data-efficient, generalizable, and memory-augmented vision-language-action framework for 3D manipulation", *arXiv preprint arXiv:2608.05042*
2. Nie Lin, Takehiko Ohkawa, Sijin Chen, Ruoshi Wen, Zhuohang Li, Liqun Huang, Zhengming Zhu, Yiming Bao, Yunfei Li, Minjie Cai, **Xiao Ma**, Wei Xu, Yoichi Sato. "SiMDex: Mining Similar Egocentric Videos for Cross-Embodiment Dexterous Manipulation", *arXiv preprint arXiv:2608.04196*
3. Zhihao Wang, Jianxiong Li, Yu Cui, Yuan Gao, Xianyuan Zhan, Junzhi Yu, **Xiao Ma**. "World Value Models for Robotic Manipulation", *arXiv preprint arXiv:2606.24742*
4. Wenbo Zhang, Jianxiong Li, Shuai Yang, Sijin Chen, Jiajun Liu, Lingqiao Liu, **Xiao Ma**. "TTT-VLA: Test-Time Latent Prompt Optimization for Vision-Language-Action Models", *arXiv preprint arXiv:2606.03127*
5. Zhuohang Li, Liqun Huang, Wei Xu, Zhengming Zhu, Nie Lin, **Xiao Ma**, Xinjun Sheng, Ruoshi Wen. "Hand-in-the-Loop: Improving VLA Policies for Dexterous Manipulation via Seamless Hand-Arm Intervention", *arXiv preprint arXiv:2605.15157*
6. Bingyi Kang*, **Xiao Ma***, Yirui Wang, Yang Yue, Shuicheng Yan. "Improving and Benchmarking Offline Reinforcement Learning Algorithms", *arXiv preprint arXiv:2306.00972* (*equal contribution)

7. Eugene Teoh*, Sumit Patidar*, **Xiao Ma**, Stephen James. “Green Screen Augmentation Enables Scene Generalisation in Robotic Manipulation”, *arXiv preprint arXiv:2407.07868* (*equal contribution)

PUBLICATIONS

1. Lei Lv, Yunfei Li, Yu Luo, Fuchun Sun, **Xiao Ma**. “FLAC: Maximum Entropy RL via Kinetic Energy Regularized Bridge Matching”. *International Conference on Machine Learning (ICML)*, 2026
2. Dong Wang, Jirong Liu, Zilong Chen, Qiao Ziqing, Xin Xiao, Bingyi Kang, Hongtao Wu, **Xiao Ma**, Tao Kong, Huaping Liu. “RoboOmni: Actions Are Just Another Modality for Your Vision-Language Models”. *International Conference on Machine Learning (ICML)*, 2026
3. Fangqi Zhu, Zhengyang Yan, Zicong Hong, Quanxin Shou, **Xiao Ma**, Song Guo, “Wmpo: World model-based policy optimization for vision-language-action models”, *International Conference on Learning Representations (ICLR)*, 2026
4. Jiawen Tian, Liqun Huang, Zhongren Cui, Jingchao Qiao, Jiafeng Xu, **Xiao Ma**, Zeyu Ren. “ByteWrist: A Parallel Robotic Wrist Enabling Flexible and Anthropomorphic Motion for Confined Spaces”, *International Conference on Robotics & Automation (ICRA)*, 2026 (oral)
5. Huaping Liu, Xinghang Li, Peiyan Li, Minghuan Liu, Dong Wang, Jirong Liu, Bingyi Kang, **Xiao Ma**, Tao Kong, Hanbo Zhang. “Towards generalist robot policies: What matters in building vision-language-action models”, *Nature Machine Intelligence (NMI)*, 2025
6. Wenbo Zhang, Tianrun Hu, Yanyuan Qiao, Hanbo Zhang, Yuchu Qin, Yang Li, Jiajun Liu, Tao Kong, Lingqiao Liu, **Xiao Ma**. “Chain-of-Action: Trajectory Autoregressive Modeling for Robotic Manipulation”, *Conference on Neural Information Processing Systems (NeurIPS)*, 2025
7. Lei Lv, Yunfei Li, Yu Luo, Fuchun Sun, Tao Kong, Jiafeng Xu, **Xiao Ma**. “Flow-Based Policy for Online Reinforcement Learning”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2025
8. Peiyan Li, Yixiang Chen, Hongtao Wu, **Xiao Ma**, Xiangnnan Wu, Yan Huang, Liang Wang, Tao Kong, Tieniu Tan. “BridgeVLA: Input-Output Alignment for Efficient 3D Manipulation Learning with Vision-Language Models”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2025
9. Wenke Xia, Yichu Yang, Hongtao Wu, **Xiao Ma**, Tao Kong, Di Hu. “Robotic Policy Learning via Human-assisted Action Preference Optimization”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2025
10. Dong Wang*, Xinghang Li*, Zhengshen Zhang, Jirong Liu, Xiao Ma, Hanbo Zhang, Tao Kong, Huaping Liu. “ProcWorld: Benchmarking Large Model Planning in Reachability-Constrained Environments”. *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025 (*equal contribution)
11. Wenbo Zhang, Yang Li, Yanyuan Qiao, Siyuan Huang, Jiajun Liu, Feras Dayoub, **Xiao Ma**, Lingqiao Liu. “Effective Tuning Strategies for Generalist Robot Manipulation Policies”, *International Conference on Robotics and Automation (ICRA)*, 2025
12. Nikita Chernyadev*, Nicholas Backshall*, **Xiao Ma***, Yunfan Lu, Younggyo Seo, Stephen James. “Bi-Gym: A Demo-Driven Mobile Bi-Manual Manipulation Benchmark”, *Conference on Robot Learning (CoRL)*, 2024 (*equal contribution)
13. Yang Yue, Bingyi Kang, **Xiao Ma**, Qisen Yang, Gao Huang, Shiji Song, Shuicheng Yan. “Decoupled Prioritized Resampling for Offline RL”, *IEEE Transactions on Neural Networks and Learning Systems, (TNNLS)*
14. **Xiao Ma**, Sumit Patidar, Iain Haughton, Stephen James. “Hierarchical Diffusion Policy for Kinematics-Aware Multi-Task Robotic Manipulation”, *Computer Vision and Pattern Recognition (CVPR)*, 2024
15. Pietro Mazzaglia, Nicholas Backshall, **Xiao Ma**, Stephen James. “Redundancy-aware Action Spaces for Robot Learning”, *IEEE Robotics and Automation Letters (RA-L)*, 2024
16. **Xiao Ma***, Bingyi Kang*, Zhongwen Xu, Min Lin, Shuicheng Yan. “Mutual Information Regularized Offline Reinforcement Learning”, *Conference on Neural Information Processing Systems (NeurIPS)*, 2023

(*equal contribution)

17. Bingyi Kang*, **Xiao Ma***, Chao Du, Tianyu Pang, Shuicheng Yan. “Efficient Diffusion Policies for Offline Reinforcement Learning”, *Conference on Neural Information Processing Systems (NeurIPS)*, 2023 (*equal contribution)
18. Jiawei Ren*, Mingyuan Zhang*, Cunjun Yu*, **Xiao Ma**, Liang Pan, Ziwei Liu. “InsActor: Instruction-driven Physics-based Characters”, *Conference on Neural Information Processing Systems (NeurIPS)*, 2023 (*equal contribution)
19. Siwei Chen, **Xiao Ma**, Zhongwen Xu. “Imitation Learning via Differentiable Physics”, *Computer Vision and Pattern Recognition (CVPR)*, 2023
20. Siwei Chen*, Cunjun Yu*, Yiqing Xu*, Linfeng Li, **Xiao Ma**, Zhongwen Xu, David Hsu. “Benchmarking Deformable Object Manipulation with Differentiable Physics”, *International Conference on Learning Representations (ICLR)*, 2023 (oral, *equal contribution)
21. Jiawei Ren*, Cunjun Yu*, Siwei Chen, **Xiao Ma**, Liang Pan, Ziwei Liu. “DiffMimic: Efficient Motion Mimicking with Differentiable Physics”, *International Conference on Learning Representations (ICLR)*, 2023 (*equal contribution)
22. Wei Qiu, **Xiao Ma**, Bo An, Svetlana Obraztsova, Shuicheng Yan, Zhongwen Xu. “RPM: Generalizable Behaviors for Multi-Agent Reinforcement Learning”, *International Conference on Learning Representations (ICLR)*, 2023
23. Hai Nguyen*, Zhihan Yang*, Andrea Baisero, **Xiao Ma**, Robert Platt, Christopher Amato. “Hierarchical Reinforcement Learning under Mixed Observability”, *Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2022 (*equal contribution)
24. **Xiao Ma**, David Hsu, Wee Sun Lee. “Learning Latent Graph Dynamics for Visual Manipulation of Deformable Objects”, *International Conference on Robotics and Automation (ICRA)*, 2022
25. Daisheng Jin*, **Xiao Ma***, Chongzhi Zhang, Yizhuo Zhou, Jiashu Tao, Mingyuan Zhang, Zhoujun Li, Xiaolong Liu. “Towards Overcoming False Positives in Visual Relationship Detection”, *British Machine Vision Conference (BMVC)*, 2021 (*equal contribution)
26. **Xiao Ma**, Siwei Chen, David Hsu, Wee Sun Lee. “Contrastive Variational Model-Based Reinforcement Learning for Complex Observations”, *The 4th Conference on Robot Learning (CoRL)*, 2020
27. Jiawei Ren, Cunjun Yu, Shunan Sheng, **Xiao Ma**, Haiyu Zhao, Shuai Yi, Hongsheng Li. “Balanced Meta-Softmax for Long-Tailed Visual Recognition”, *Advances in Neural Information Processing Systems (NeurIPS)*, 2020
28. Siwei Chen, **Xiao Ma**, David Hsu. “DinerDash Gym: A Benchmark for Policy Learning in High-Dimensional Action Space”, *In IL workshop, Robotics: Science and Systems (RSS)*, 2020
29. Cunjun Yu*, **Xiao Ma***, Jiawei Ren, Haiyu Zhao, Shuai Yi. “Spatio-Temporal Graph Transformer Networks for Pedestrian Trajectory Prediction”, *European Conference on Computer Vision (ECCV)*, 2020 (*equal contribution)
30. Zuowu Zheng, Xiaofeng Gao, **Xiao Ma**, Guihai Chen. “Predicting Hot Events in the Early Period through Bayesian Model for Social Networks”, *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2020
31. **Xiao Ma**, Peter Karkus, David Hsu, Wee Sun Lee, Nan Ye. “Discriminative Particle Filter Reinforcement Learning for Complex Partial Observations”, *International Conference on Learning Representations (ICLR)*, 2020
32. **Xiao Ma***, Peter Karkus*, David Hsu, Wee Sun Lee. “Particle Filter Recurrent Neural Networks”, *AAAI Conference on Artificial Intelligence (AAAI)*, 2020 (*equal contribution)
33. Peter Karkus, **Xiao Ma**, David Hsu, Leslie Pack Kaelbling, Wee Sun Lee, Tomas Lozano-Perez. “Differentiable Algorithm Networks for Composable Robot Learning”, *Robotics: Science and Systems (RSS)*, 2019 (Best Student/System Paper Finalist)

34. **Xiao Ma**, Peter Karkus, David Hsu, Wee Sun Lee “PF-LSTM: Belief State Particle Filter for LSTM”, *In RLPO Workshop, Advances in Neural Information Processing Systems (NeurIPS)*, 2018
35. **Xiao Ma**, Xiaofeng Gao, Guihai Chen. “BEEP: a Bayesian perspective Early state Event Prediction model for online social networks”, *IEEE International Conference on Data Mining (ICDM)*, 2017
36. **Xiao Ma**, Zhenzhe Zheng, Fan Wu and Guihai Chen. “Trust-Based Time Series Data Model for Mobile Crowdsensing”, *IEEE International Conference on Communications (ICC)*, 2017

PROFESSIONAL ACTIVITIES

Area Chair

- Conferences: ICLR (2025, 2026), NeurIPS (2026), ICML (2026), CoRL (2026)

Reviewer

- Conferences: NeurIPS (2020 - 2025), WACV (2021 - 2023), ICRA (2021 - 2025), AAAI (2021 - 2025), CVPR (2021 - 2023), ICLR (2022 - 2026), ICML (2022 - 2025)
- Journals: RA-L (2021 - 2025)